Bluetooth Controlled Home Automation

1. Introduction

Home automation is the use of electronic and communication technologies to control household devices automatically or remotely.

This project, **Bluetooth Controlled Home Automation**, demonstrates how a smartphone can be used to wirelessly control electrical devices using Bluetooth communication.

An **Arduino UNO** acts as the main controller. An **HC-05 Bluetooth module** receives commands from a smartphone and sends them to the Arduino. The Arduino processes these commands and controls a relay module. For a safe beginner demonstration, LEDs or other low-voltage loads can be connected to the relay outputs.

2Problem. Statement

Controlling electrical devices manually can sometimes be inconvenient. A simple wireless system can make it easier to control devices from a smartphone.

This project provides a basic solution by allowing the user to send ON/OFF commands through Bluetooth.

. 3Objectives

The main objectives of this project are:

1. To understand the basics of home automation.
2. To learn how Arduino UNO works as a microcontroller.
3. To understand Bluetooth communication.
4. To control devices wirelessly using a smartphone.
5. To understand the operation of relay modules.
6. To develop basic embedded-system programming skills.
7. To create a simple ECE project that can be expanded in the future.

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4. Components Required

Component	Quantity	Purpose
Arduino UNO	1	Main controller

Component	Quantity	Purpose
HC-05 Bluetooth Module	1	Wireless communication
2-Channel Relay Module	1	Device switching
LED	2	Low-voltage demonstration loads
Resistor	2	LED current limiting
Breadboard	1	Circuit prototyping
Jumper Wires	As required	Connections
Smartphone	1	Sending Bluetooth commands
USB Cable	1	Programming Arduino

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5. Block Diagram

```
      Smartphone
           |
           | Bluetooth
           ↓
      HC-05 Module
           |
           | Serial Communication
           ↓
      Arduino UNO
           |
       ┌───┴───┐
       ↓       ↓
   Relay 1   Relay 2
       ↓       ↓
     LED 1    LED 2
```

---

6. Working Principle

The system works through serial communication between the HC-05 Bluetooth module and Arduino UNO.

Step 1 — Smartphone

A Bluetooth terminal/controller application is used on the smartphone.

The user sends a command such as:

```
1
```

or

```
0
```

Step 2 — HC-05 Bluetooth Module

The HC-05 receives the command wirelessly from the smartphone.

It transfers the received character to the Arduino through serial communication.

Step 3 — Arduino UNO

Arduino checks the received character.

For example:

- `1` means Device 1 ON.
- `0` means Device 1 OFF.
- `2` means Device 2 ON.
- `3` means Device 2 OFF.

Step 4 — Relay Module

The Arduino changes the corresponding relay output.

The relay acts as an electrically controlled switch.

For this beginner demonstration, LEDs can be used instead of household AC appliances.

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7. Command Table

Command	Function
`1`	Device 1 ON
`0`	Device 1 OFF
`2`	Device 2 ON
`3`	Device 2 OFF

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8. Circuit Connections

HC-05 to Arduino UNO

HC-05	Arduino
VCC	5V
GND	GND
TXD	Arduino software serial RX
RXD	Arduino software serial TX

Relay Module to Arduino

Relay	Arduino
VCC	5V
GND	GND
IN1	Digital Pin 7
IN2	Digital Pin 8

The exact relay input behavior can vary by module. Some relay boards are **active LOW**, while others are **active HIGH**. The code below assumes an active-LOW relay module.

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9. Arduino Program

The program uses the `SoftwareSerial` library so that the normal USB serial connection can still be used for debugging.

```
#include <SoftwareSerial.h>

// HC-05 connection
// Arduino pin 10 = RX
// Arduino pin 11 = TX
SoftwareSerial Bluetooth(10, 11);

// Relay pins
const int relay1 = 7;
const int relay2 = 8;

void setup() {
  // Start serial communication
```

```
  Serial.begin(9600);
  Bluetooth.begin(9600);

  // Set relay pins as outputs
  pinMode(relay1, OUTPUT);
  pinMode(relay2, OUTPUT);

  // Turn both devices OFF initially
  digitalWrite(relay1, HIGH);
  digitalWrite(relay2, HIGH);

  Serial.println("Bluetooth Home Automation Started");
  Serial.println("Send 1, 0, 2 or 3");
}

void loop() {

  // Check whether Bluetooth has received a command
  if (Bluetooth.available()) {

    char command = Bluetooth.read();

    Serial.print("Received command: ");
    Serial.println(command);

    // Device 1 ON
    if (command == '1') {
      digitalWrite(relay1, LOW);
      Serial.println("Device 1 ON");
    }

    // Device 1 OFF
    else if (command == '0') {
      digitalWrite(relay1, HIGH);
      Serial.println("Device 1 OFF");
    }

    // Device 2 ON
    else if (command == '2') {
      digitalWrite(relay2, LOW);
      Serial.println("Device 2 ON");
    }

    // Device 2 OFF
    else if (command == '3') {
      digitalWrite(relay2, HIGH);
      Serial.println("Device 2 OFF");
    }
```

```
  }
}
```

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10. Code Explanation

Library

```
#include <SoftwareSerial.h>
```

This library allows Arduino to communicate with the HC-05 through digital pins.

Bluetooth Object

```
SoftwareSerial Bluetooth(10, 11);
```

Here:

- Pin 10 is used as Arduino RX.
- Pin 11 is used as Arduino TX.

Relay Pins

```
const int relay1 = 7;
const int relay2 = 8;
```

These pins control the two relay channels.

Setup

```
pinMode(relay1, OUTPUT);
pinMode(relay2, OUTPUT);
```

This tells Arduino that the relay pins will be used as outputs.

Reading Bluetooth Data

```
if (Bluetooth.available()) {
    char command = Bluetooth.read();
}
```

Arduino checks whether a Bluetooth command has arrived and reads it as a character.

Controlling the Relay

```
if (command == '1') {
    digitalWrite(relay1, LOW);
}
```

When the user sends `1`, Relay 1 is activated.

Similarly:

```
0 → Relay 1 OFF
2 → Relay 2 ON
3 → Relay 2 OFF
```

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11. How to Upload the Code

1. Install Arduino IDE.
2. Connect Arduino UNO to the computer using USB.
3. Open the Arduino IDE.
4. Create a new sketch.
5. Paste the program.
6. Select **Arduino UNO** as the board.
7. Select the correct COM port.
8. Click **Verify**.
9. Click **Upload**.
10. After uploading, connect the Bluetooth module according to the circuit.

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12. Smartphone Control

A Bluetooth terminal/controller application can be used to send commands.

First pair the smartphone with the HC-05 module.

Then send:

```
1
```

to turn Device 1 ON.

Send:

```
0
```

to turn Device 1 OFF.

Send:

```
2
```

to turn Device 2 ON.

Send:

```
3
```

to turn Device 2 OFF.

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13. Expected Output

When the user sends a valid command, the Arduino performs the corresponding operation.

Example:

```
Command: 1
Device 1 ON

Command: 0
Device 1 OFF

Command: 2
Device 2 ON

Command: 3
Device 2 OFF
```

---

14. Advantages

- Simple and beginner-friendly.

- Wireless control.
- Low-cost components.
- Easy to program.
- Useful for learning embedded systems.
- Can be expanded with additional devices.
- Demonstrates communication between a smartphone and microcontroller.

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15. Limitations

- Bluetooth range is limited.
- The system generally requires the smartphone to remain within Bluetooth range.
- It does not provide internet-based remote control.
- The number of controllable devices depends on the available controller outputs and hardware.
- Relay modules may require additional considerations depending on the load.

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16. Future Scope

The project can be improved by adding:

IoT Control

An ESP32 or Wi-Fi-enabled controller can be used to control devices through the internet.

Voice Control

Voice commands can be added using a suitable mobile application or cloud-based system.

Sensors

Sensors such as:

- Temperature sensor
- PIR motion sensor
- LDR
- Gas sensor

can be added to make the system more intelligent.

Mobile Application

A dedicated Android application can provide buttons for each appliance instead of manually entering commands.

Energy Monitoring

Current and voltage sensors can be added to monitor energy consumption.

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17. Applications

This concept can be used in:

- Smart homes
- Small automation systems
- Educational laboratories
- Demonstration projects
- Assistive technology
- Basic IoT learning

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18. Learning Outcomes

After completing this project, a beginner can understand:

- Arduino UNO basics
- Digital input and output
- Serial communication
- Bluetooth communication
- Relay operation
- Basic C/C++ programming
- Embedded-system concepts
- GitHub project documentation

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19. Conclusion

The **Bluetooth Controlled Home Automation** project demonstrates how wireless communication can be combined with a microcontroller to control devices.

The HC-05 Bluetooth module provides communication between the smartphone and Arduino UNO, while the Arduino processes the received commands and controls the relay module.

This project provides a simple foundation for learning **embedded systems, electronics, communication, and home automation**, making it suitable for an ECE beginner.

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20. Safety Note

For a beginner demonstration, use LEDs or other appropriate low-voltage loads.

Household AC mains can cause serious electric shock, fire, or equipment damage. Do not experiment with exposed mains wiring. If actual AC appliances are to be controlled, the installation should be designed and connected using appropriate rated components and handled by a qualified person.